sns_app.py

```
import streamlit as st
import seaborn as sns
import matplotlib.pyplot as plt

sns.set_theme(style="whitegrid")

tips = sns.load_dataset("tips")

st.title('Mr.Raghu Sai Varun seaborn bootcamp tips data visualization app')
st.write("This is a simple app to visualize the tips dataset using seaborn.")

# function create and display plot
def display_plot(title, plot_func):
    st.subheader(title)
    fig, ax = plt.subplots(figsize=(8, 6))
    plot_func(ax=ax)
    st.pyplot(fig)
    plt.close(fig)

# plot

def scatter_plot(ax):
    sns.scatterplot(data=tips, x="total_bill", y="tip",hue="time", size="size",
palette="deep",ax=ax)
    ax.set_title("Scatter plot of total bill vs tip")

def line_plot(ax):
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    sns.lineplot(data=tips, x= 'size', y='total_bill', hue='sex',markers='o',ax=ax)
    ax.set_title("Line plot of total bill vs tip")

def bar_plot(ax):
    sns.barplot(data=tips, x='day', y='total_bill', hue = 'sex',palette='muted',ax=ax)
    ax.set_title("Barplot of Total Bill by Day")

def box_plot(ax):
    sns.boxplot(data=tips, x='day', y='tip', hue='smoker', palette='Set2',ax=ax)
    ax.set_title("Boxplot of Tips by Day and Smoker Status")

def violin_plot(ax):
    sns.violinplot(data=tips, x='day', y='total_bill', hue='time', split=True,
palette='pastel',ax=ax)
    ax.set_title("Violin Plot of Total Bill by Day and Time")

def count_plot(ax):
    sns.countplot(data=tips, x='day', hue='smoker', palette='dark',ax=ax)
    ax.set_title("Count Plot of Days by Smoker Status")

def reg_plot(ax):
    sns.regplot(data=tips, x='total_bill', y='tip', scatter_kws={'s':50},
line_kws={'color':'red'},ax=ax)
    ax.set_title("Regression Plot of Total Bill vs Tip")

def hist_plot(ax):
    sns.histplot(data=tips, x='total_bill', bins=20, kde=True, color='blue',ax=ax)
    ax.set_title("Histogram of Total Bill with KDE")
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def strip_plot(ax):
    sns.stripplot(data=tips, x='day', y='tip', hue='sex', jitter=True, palette='Set1',ax=ax)
    ax.set_title("strip plot: Tips by data and gender")

def kde_plot(ax):
    sns.kdeplot(data=tips, x='total_bill',hue='sex', fill=True, palette='tab10',ax=ax)
    ax.set_title("kde plot:Total bill density by gender")


display_plot("Scatter Plot", scatter_plot)
display_plot("Line Plot", line_plot)
display_plot("Bar Plot", bar_plot)
display_plot("Box Plot", box_plot)
display_plot("Violin Plot", violin_plot)
display_plot("Count Plot", count_plot)
display_plot("Regression Plot", reg_plot)
display_plot("Histogram Plot", hist_plot)
display_plot("Strip Plot", strip_plot)
display_plot("KDE Plot", kde_plot)
```
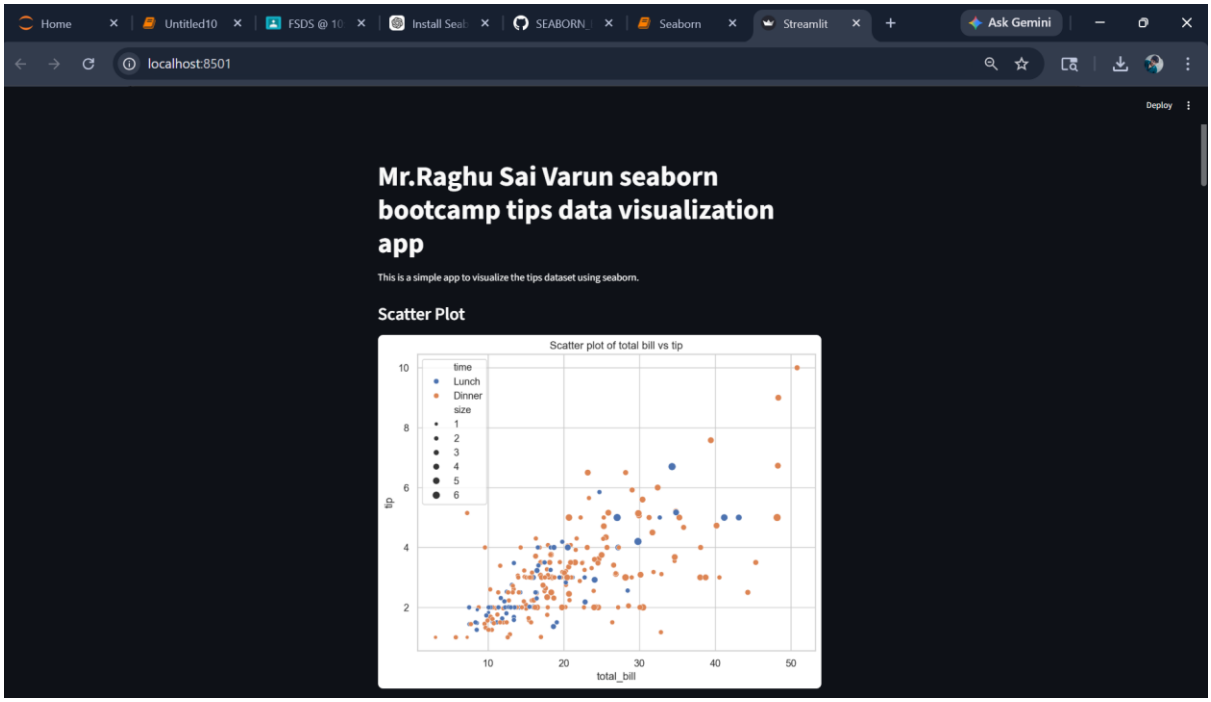
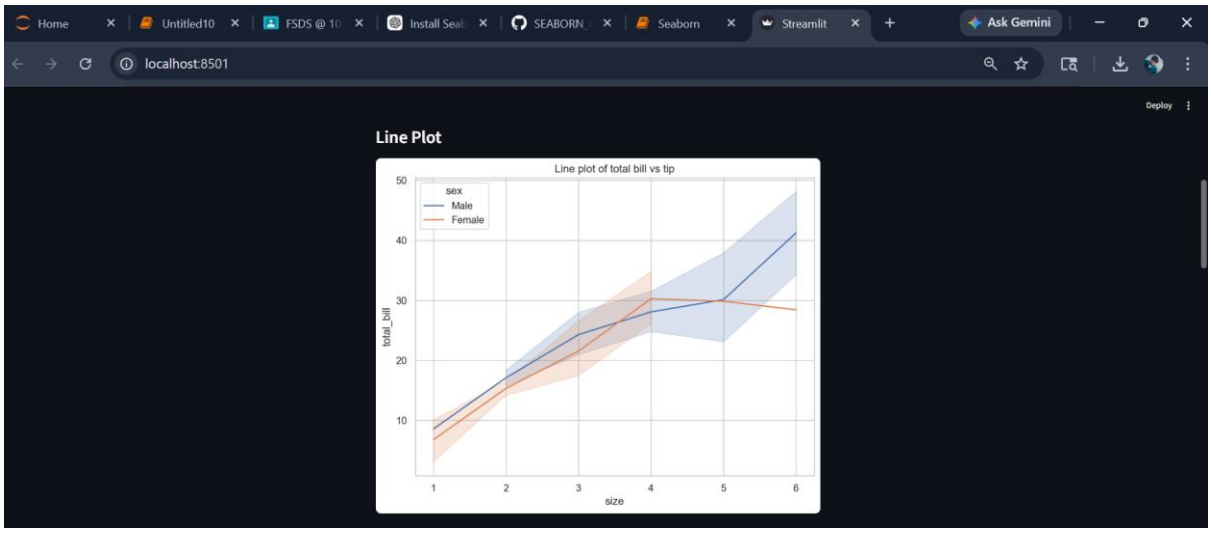
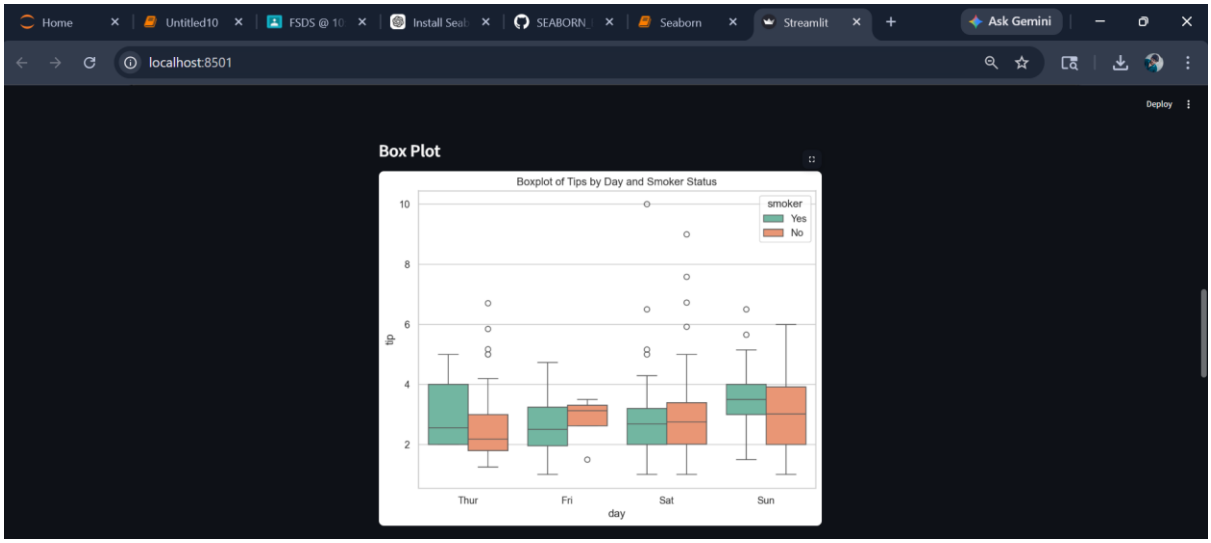
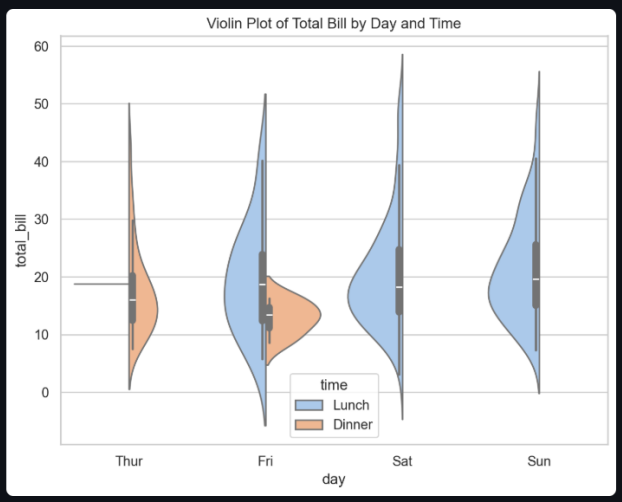


Home
Untitled10
FSDS @ 10
Install Seab
SEABORN_
Seaborn
Streamlit
Ask Gemini
localhost:8501
Deploy
Mr.Raghu Sai Varun seaborn bootcamp tips data visualization app
This is a simple app to visualize the tips dataset using seaborn.
Scatter Plot
Scatter plot of total bill vs tip
time
Lunch
Dinner
size
1
2
3
4
5
6
tip
total_bill

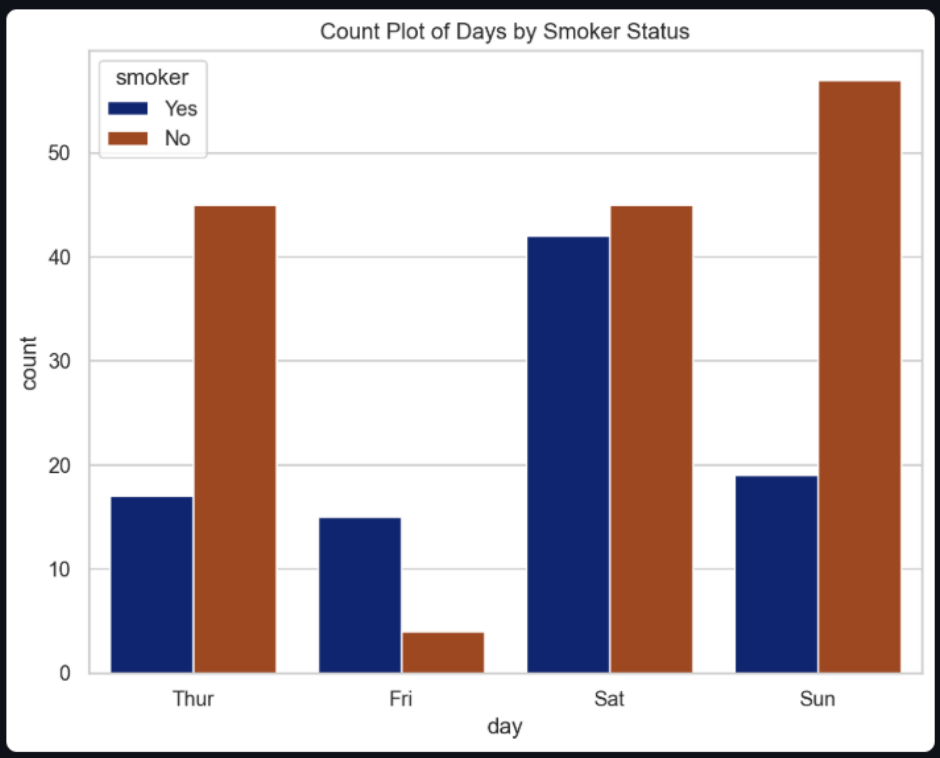

Home
Untitled10
FSDS @ 10
Install Seab
SEABORN_
Seaborn
Streamlit
Ask Gemini
localhost:8501
Deploy
Line Plot
Line plot of total bill vs tip
sex
Male
Female
total_bill
size


Home
Untitled10
FSDS @ 10
Install Seab
SEABORN_
Seaborn
Streamlit
Ask Gemini
localhost:8501
Deploy
Box Plot
Boxplot of Tips by Day and Smoker Status
smoker
Yes
No
tip
Thur
Fri
Sat
Sun
day

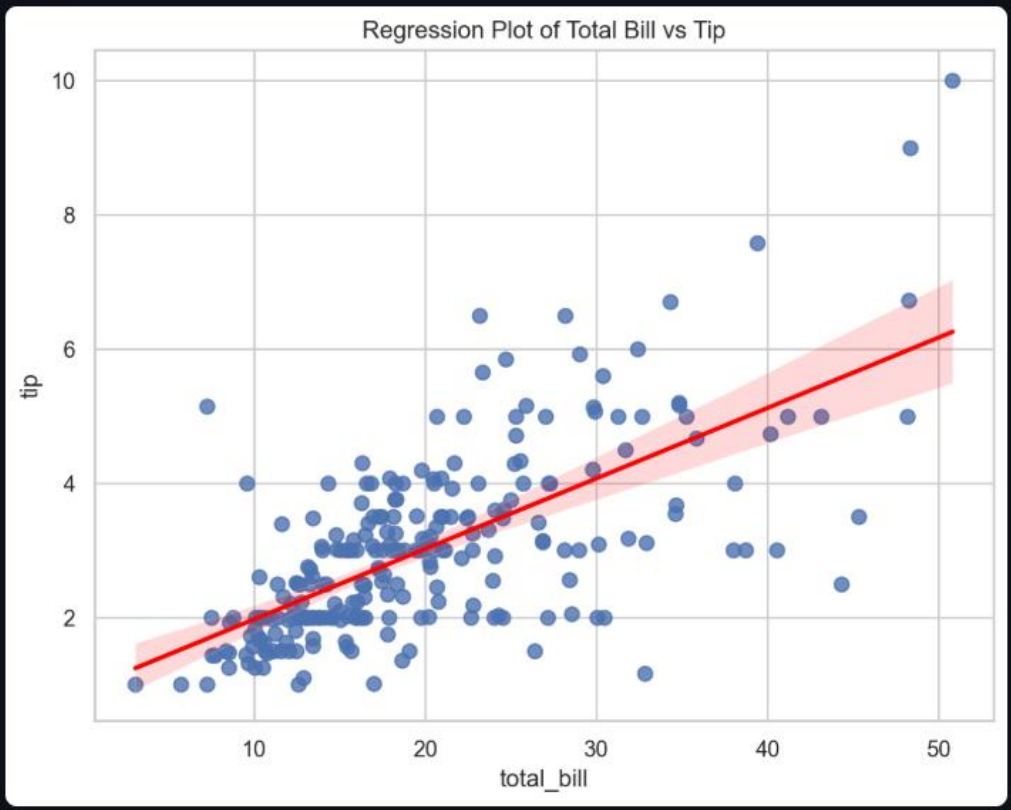
Violin Plot



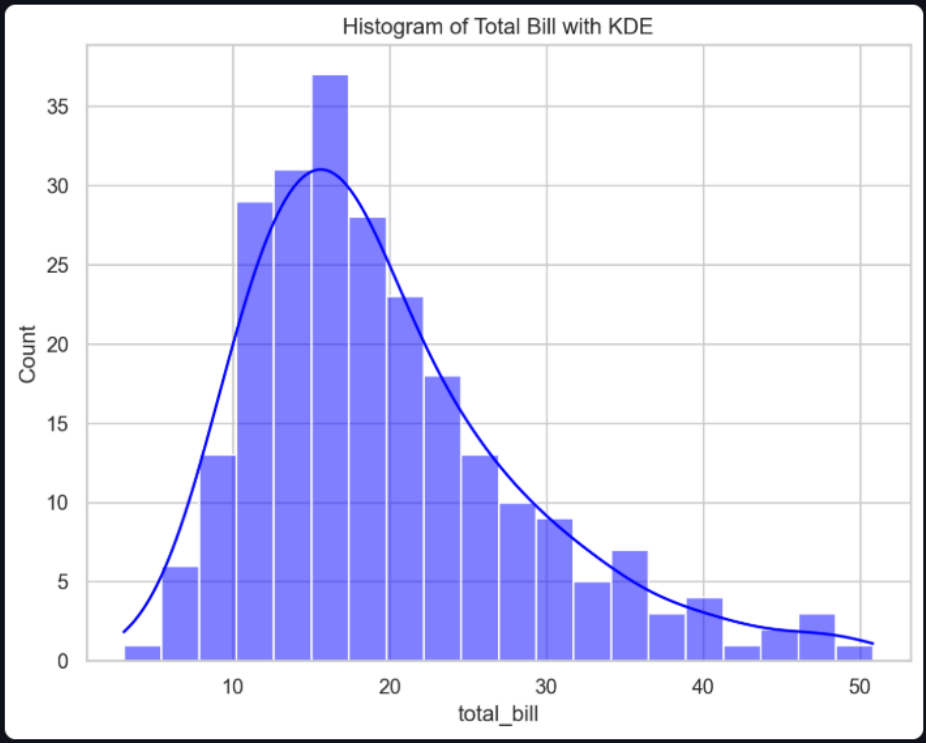
Count Plot



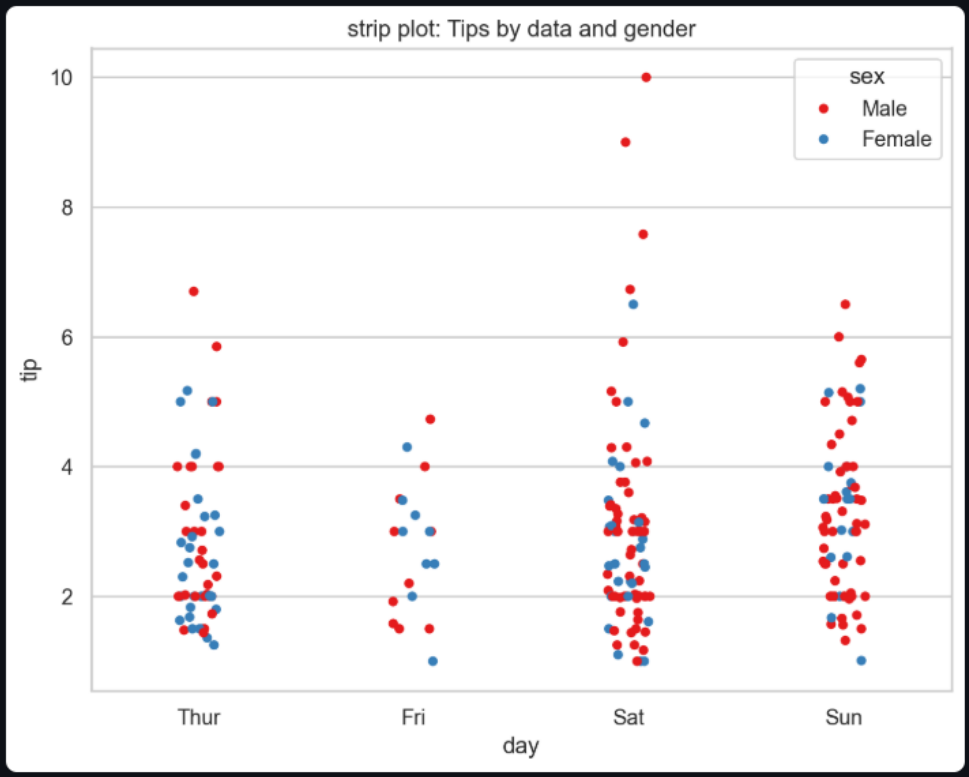
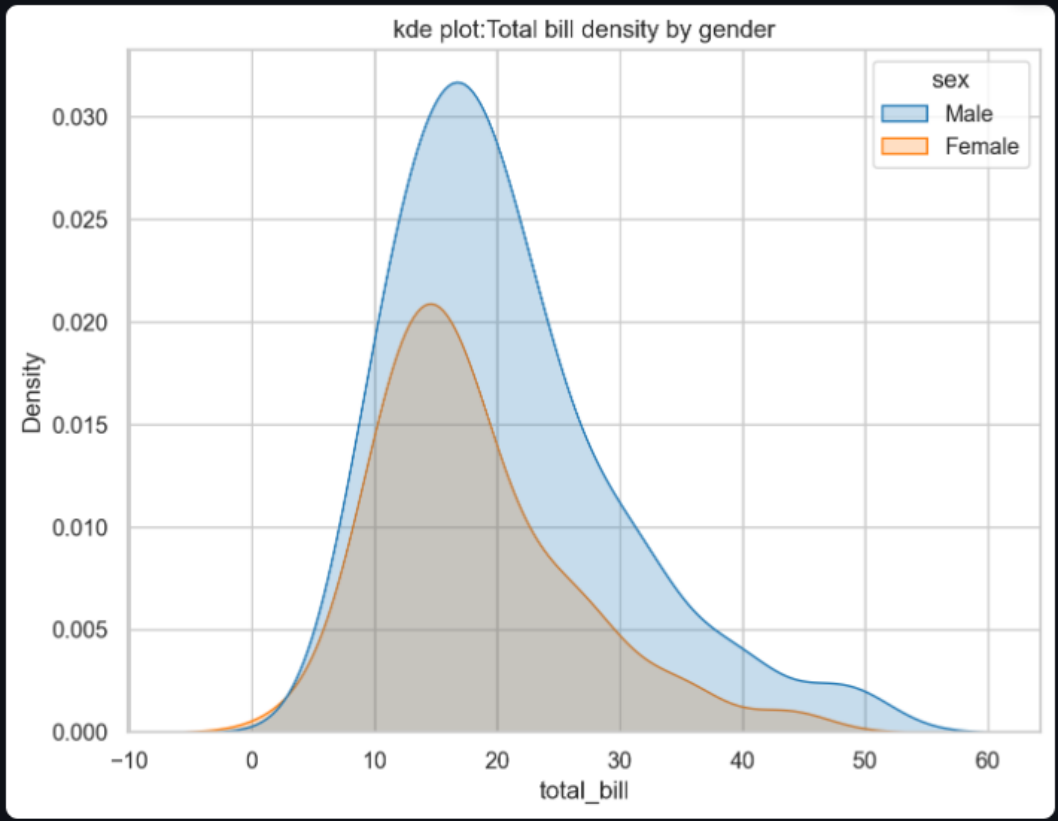
Regression Plot



Histogram Plot



Strip Plot



KDE Plot